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ECON 265 Quantitative Microeconomics

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Investigating the Impact of the 2025 Trade Wars on US Domestic Prices

1. Introduction

It seems that all anyone can talk about nowadays is the United States' (US) trade policy. Since the commencement of Trump's second administration, US trade policy has been defined by an escalating tariff onslaught against American trade partners. These tariffs have increased input costs and contributed to supply chain disruptions across various industries. The new landscape of US trade policy, characterized by ongoing trade disputes and policy changes, has renewed interest in understanding the impact of tariffs on prices paid by US consumers. Economists have debated the extent to which tariffs affect domestic prices, with many economists' initial projections of long term above-average inflation seemingly unfounded, as the US core inflation rate stands at 2.6% as of March 2026. The inflation rate fails to capture the true influence of tariffs on prices, as industries are asymmetrically affected, resulting in the moderate price increases observed in the US economy. Through a case study of the automobile industry, specifically the effect of the 2025 tariff on foreign cars and car parts, this paper investigates the extent to which tariffs affect the prices domestic consumers receive in affected industries such as the automobile industry.

In the year prior to when the tariff went into effect, the US imported \$219.5 billion in passenger vehicles, up about a 4.4% from the year before (Ullman, 2025, par. 2). This trend reflects the increasing demand from US consumers for foreign-manufactured vehicles, making tariffs a particularly relevant issue for the automobile industry. According to traditional microeconomic theory, a tariff acts as a tax on imports, reducing consumer surplus and producer surplus (i.e., increasing consumers' prices and decreasing producers' profits).

The pass-through rate measures the extent to which increased costs are passed to consumers (i.e. who bears the incidence) and depends on market conditions and the elasticity of demand. If the consumer demand for a good is inelastic, the pass-through of increased tariff-related costs to consumers is usually high. However, when consumers can easily switch to a cheaper domestic version, indicating consumer demand for the good is elastic, the pass-through to consumers is usually low. To avoid losing their customers, firms may absorb part of the tariff burden, effectively lowering their profit margins and reducing price increases.

The Trump administration explains that the tariffs plaguing the automobile industry are a tool to address the decline in manufacturing output, which has reduced U.S. manufacturing capacity. However, the US automobile sector trade deficit hit \$160.3 billion, signaling continued reliance on foreign-made vehicles even as domestic output grows (Ullman, 2025, par. 2). This reliance on foreign vehicles by American consumers suggests relatively high demand from domestic consumers for foreign-made vehicles, despite domestic alternatives, so there's a low likelihood that foreign producers will bear the cost of tariffs, leaving consumers to bear them completely. In addition, US automobile manufacturers face higher costs, as integral car parts are sourced internationally. (Lincicome and Haven, 2025, par. 6) Tariffs will increase input costs for domestic automobile manufacturers, raising the prices of domestic automobiles. The relatively strong elasticity of demand for domestic automobiles may cause domestic manufacturers to absorb the burden, thereby counteracting the expected complete pass-through of tariff increases into new-vehicle prices.

This paper examines the extent to which tariffs on imported automobiles and car parts are passed through to U.S. consumers through changes in vehicle prices. Since pass-through reflects the extent to which consumers bear the burden of tariffs, the results provide indirect evidence on consumer tax incidence. Using a difference-in-differences model, this paper analyzes the impact of tariffs on imported cars and parts on US households by examining price changes in new and used vehicles after the date the tariff took effect. The paper uses seasonally adjusted Consumer Price Index (CPI) data published by the US Bureau of Labor Statistics

(BLS), which measures average price changes for new vehicles, used cars, and trucks. The main finding of this study is that prices for new vehicles decreased following the imposition of tariffs on imported cars and car parts, suggesting incomplete pass-through of tariffs into new vehicle consumer prices. A secondary finding of this study is that prices for used vehicles increased after the tariff took effect, suggesting the presence of a hidden mechanism, substitution effects within the market.

Literature Review

Most of the literature on tariffs' effects on prices in the US centers on the 2018-2019 tariffs levied against China during the first Trump administration. Contemporary literature has expanded in response to the new landscape of US trade policy during the second Trump administration, which has seen tariffs extend beyond goods from China to affect a wider range of goods from a majority of the US's trading partners. Existing research largely examines the extent to which tariffs are passed through into domestic prices and the implications of higher import costs for the domestic economy. Amiti, Redding, and Weinstein (2019) provide much of the seminal literature on the impact of tariffs on prices and welfare during the 2018-2019 period and include an extension examining retaliatory tariffs. Subsequent work focusing on the 2018-2019 tariffs on China has expanded to examine the terms of trade and employ empirical models to uncover the pass-through of tariffs into consumer prices.

Amiti, Redding, and Weinstein (2019) use the 2018-2019 tariffs as a natural experiment to illustrate how conventional trade models provide a strong conceptual framework for understanding how tariffs affect prices, trade volumes, and welfare. They estimate the pass-through of tariffs to prices through changes in the prices paid by US importers (Amiti, Redding, and Weinstein, 2019, pg. 193). They find that US import tariffs were almost completely passed through to US domestic prices in 2018, implying that much of the incidence of the tariffs fell on domestic consumers and importers, with little impact on the prices received by foreign exporters (Amiti, Redding, and Weinstein, 2019, pg. 195). Pfajgelbaum, Goldberg, Kennedy, and Khandelwal

(2019) expand on this literature by estimating key elasticities of trade outcomes using import and retaliatory tariff variation and embedding the estimates into a quantitative general-equilibrium trade model of the U.S. economy (Pfajgelbaum et al., 2019, pg.2). Their analysis relies on detailed trade data, including bilateral import and export flows, tariff schedules, and industry expenditure shares during the 2018–2019 tariff period (Pfajgelbaum et al., 2019, pg. 7). They conclude that the prices of imports subject to tariffs did not fall, implying near-complete pass-through of tariffs into prices (Pfajgelbaum et al., 2019, pg.52). Their findings further reinforce the broader literature’s conclusion that domestic consumers bear the bulk of the burden of tariff-related price increases. Cavallo, Gopinath, Neiman, and Tang (2021) analyze disaggregated microdata on exchange rates, import prices (ex-tariff), and retail prices to assess the effects of tariffs and exchange rate movements on import prices (Cavallo et al., 2021, pg. 20). Similar to Amiti, Redding, and Weinstein (2019), they find that tariffs are passed through almost fully into US import prices (Cavallo et al., 2021, pg. 32), implying that much of the incidence of tariffs rests with US consumers and importers. Using the same data, they find substantially lower pass-through from exchange-rate shocks into import prices, suggesting that the depreciation of the Chinese renminbi against the US dollar during the summer of 2019 did little to offset the impact of the tariffs (Cavallo et al., 2021, pg. 32),. Their methodology is particularly significant because it traces tariff pass-through across multiple stages of the pricing chain, from importers to retailers.

Contrary to much of the broader tariff literature, Ahmad and Ahmad’s (2022) work traces the effects of tariffs using the steel industry as a case study. Their decision to investigate the steel industry in isolation stems from Amiti, Redding, and Weinstein’s (2020) findings that steel products, particularly steel inputs, behaved differently from many other imports during the tariff period (Ahmad & Ahmad, 2022, 482). They report that tariffs on steel inputs are borne by exporters (Amiti, Redding, and Weinstein, 2020). This study, like mine, contributes to the literature by analyzing the impacts of tariffs on prices within a specific industry. Ahmad and Ahmad (2022) use an event-study regression model to analyze the impact of U.S. tariffs on steel import values

and unit prices. Their data comes from the U.S. Customs Department and includes products at the HTS 10-digit and country levels, with monthly import values and volumes (Ahmad & Ahmad, 2022, 486). They find substantial tariff pass-through into import prices in both the short and long term, although exporters absorb part of the burden through price reductions over time (Ahmad & Ahmad, 2022, 490). They additionally find that tariffs contributed to increased domestic steel production. However, supply chain adjustments and import diversification towards other countries became significant in the long term (Ahmad & Ahmad, 2022, 491-2).

In its 2025 publication, the Federal Reserve Board of Governors proposes a novel methodology that uses inflation measures and import shares (the Fed is partial to using Personal Consumption Expenditures) to detect the effects of tariffs on prices in real time (Minton & Somale, 2025, par. 2). Economists interested in differentiating the effects of tariffs from broader macroeconomic factors have struggled because existing methods fail to adequately assess tariffs' effects on consumer prices in real time (Minton & Somale, 2025, par. 1). Their framework assumes full direct pass-through, where the prices of affected imports rise in proportion to tariffs while import volumes remain constant (Minton & Somale, 2025, par. 8). This methodology attempts to isolate tariff-driven inflationary pressures from broader macroeconomic movements in real time using aggregate inflation data. Using an event-study approach within their theoretical framework to assess the effects of the 2018-2019 tariffs, the authors find that tariffs were fully and quickly passed through to consumer prices (Minton & Somale, 2025, par. 23).

This paper contributes to the literature by examining the automobile industry specifically and analyzing how the 2025 tariffs on imported automobiles and automobile parts affected the prices of new and used vehicles in the United States. Using Consumer Price Index data and a difference-in-differences framework, this paper extends the existing literature by examining consumer-level price responses within a specific domestic market, rather than relying solely on broader aggregate inflation measures or import prices. The paper's findings also

suggest that substitution within the automobile market may help explain the limited pass-through of tariffs into new-vehicle prices.

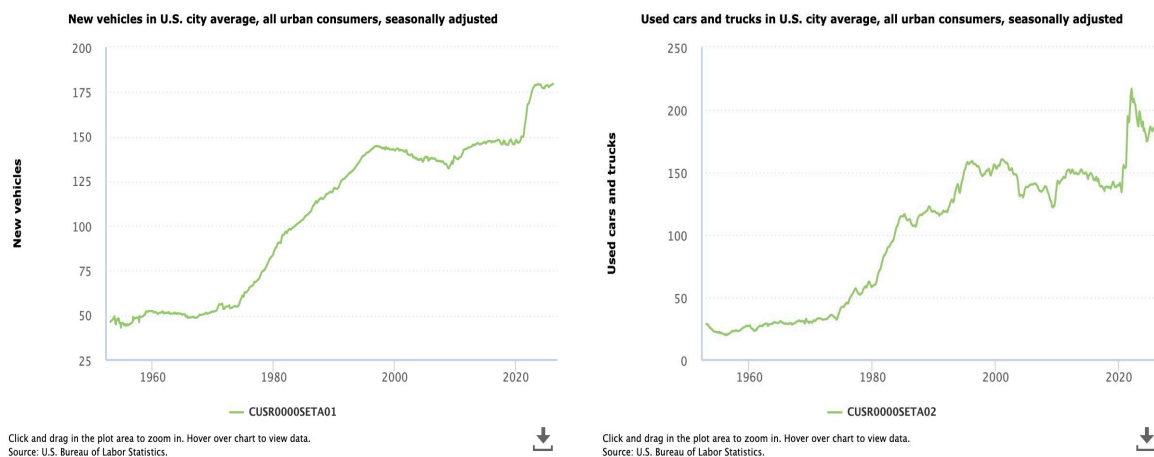
2. Data

This paper uses aggregated, seasonally adjusted Consumer Price Index (CPI) data published monthly by the Bureau of Labor Statistics (BLS) and hosted on the Federal Reserve Economic Data site. CPI measures the average change over time in the prices consumers pay for a representative basket of consumer goods and services and is commonly used as a measure of inflation experienced by consumers. The particular CPI datasets used in this paper are the broadest and most comprehensive CPI, called the All Items Consumer Price Index for All Urban Consumers for the U.S. City Average. This measure uses 1982-84 as the base year; the index represents the corresponding price level “100”. For clarity, an index report of 50 indicates that the current price of that item is 50% lower than it was during the designated base period. An index report of 150 represents a 50% increase in price compared to the designated base period. It’s important to note that the index represents changes in the average price of goods compared to the base year, rather than specific, individual vehicle purchase details. The used cars and trucks index comprises used vehicles aged 2 to 7 years. Subcompact, compact, sporty, intermediate, full, and luxury cars are all included in the index. The CPI new vehicles index measures price changes for newly produced cars and trucks (current or previous model year) that have not been previously titled. It covers consumer vehicles, including subcompact to luxury cars, pickups, vans, and SUVs, utilizing transaction-level data to track price changes.

The empirical analysis uses seasonally adjusted CPI data for new vehicles and used cars and trucks for all urban consumers in the U.S. city average. The used cars and trucks index serves as the control group, while the new vehicles index serves as the treatment group. The BLS publishes both seasonally adjusted and non-seasonally adjusted CPI series. This paper uses seasonally adjusted data to remove predictable annual price

fluctuations, such as seasonal demand shifts, allowing for a clearer analysis of short-term price movements following tariff implementation.

Several data-cleaning and transformation procedures were conducted before analysis. First, monthly date variables were constructed from the reported year and period variables. Second, CPI index values were converted to numeric form. Finally, the CPI indices were logged before estimation. Logging the CPI values converts absolute CPI levels into a proxy for percentage changes, allowing the regression coefficients to be interpreted as approximately percentage changes in prices rather than as changes in index points.



Figures 1 and 2 plot the logged monthly CPI indices for used cars and trucks and new vehicles from January 1953 through March 2026. Both series display clear positive trends over time. However, several major macroeconomic events produced noticeable deviations from these broader trends. In the used vehicle series, prices declined during the 2008–2009 Great Financial Crisis. Prices rose sharply during the COVID-19 pandemic, as supply-chain disruptions stemming from semiconductor shortages severely constrained vehicle supply while demand remained unexpectedly high. These market dynamics influenced the rise in used vehicle prices. Similar patterns appear in the new vehicle series. New-vehicle prices showed a positive trend until the late 1990s, through the beginning of the Great Financial Crisis. The price downturn was partly due to increased competition from foreign manufacturers and to manufacturers' reliance on incentives to attract buyers.

Following the Great Financial Crisis, new vehicle prices began rising more rapidly due to rising manufacturing costs, increased demand for larger SUVs and trucks, and the incorporation of advanced safety and technology features. Prices increased sharply again following the COVID-19 pandemic as supply-chain disruptions and semiconductor shortages constrained production while consumer demand remained elevated.

3. Empirical Model

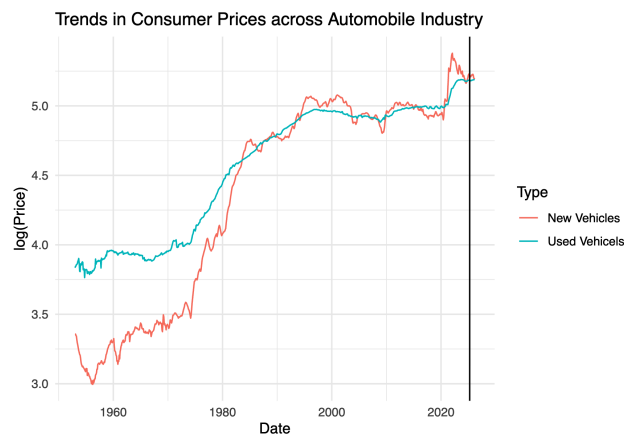
To analyze the effect of the 25% tariffs on imported automobiles and specific automobile parts under the Trump administration, which took effect on April 3, 2025, on prices, this paper employs a difference-in-differences model using CPI data for used cars and trucks and new vehicles from January 1953 through March 2026. The model tests whether the prices of new vehicles change in response to the 25% tariff relative to old cars and trucks, controlling for time fixed effects. The model is as follows:

$$\log(\text{Price}_{it}) = \beta_0 + \beta_1 \text{Treat}_i + \beta_2 \text{Post}_t + \beta_3 (\text{Treat}_i \times \text{Post}_t) + \gamma_t + u_{i,t} \quad (1)$$

where the dependent variable, $\log(\text{Price}_{it})$, measures the logged CPI price index for vehicle category i in time period t . Treat_i is a binary variable equal to 1 if the vehicle category is designated as a new vehicle and 0 if it is part of the control group (used cars and trucks). Post is also a binary variable that equals 1 if the time period is after April 2025, and 0 otherwise. The parameter γ_t is the time fixed effects, which control for broader domestic shocks and time trends affecting both groups simultaneously. The parameter $u_{i,t}$ is the error term, which captures components of the model that the independent variables fail to explain. The coefficient of interest, β_3 , represents the effect of tariffs on the logged CPI price index for new vehicles. The coefficient β_2 represents the post-tariff trend for both groups, while β_1 represents the average difference between new and used

vehicle prices before tariffs. The coefficient β_0 corresponds to the intercept and represents the average logged CPI for used vehicles before tariffs.

This empirical strategy relies on the assumption of parallel pre-trends in the CPI data for new vehicles and used cars and trucks, implying that, in the absence of tariffs, the CPI price levels for new vehicles would have followed similar trends to those of the control group. In selecting used cars as the control group, this paper assumes that tariffs would not directly affect vehicles containing car parts already circulating within the domestic market. In addition, used vehicles are part of the broader automobile market, so broader shocks to the domestic automobile industry should be reflected symmetrically in both groups.



To evaluate the plausibility of the parallel trends assumption, this paper plots the logged CPI indices for new vehicles, used cars, and trucks over time. The data display imperfect but reasonably similar pre-treatment trends in the years immediately preceding the implementation of the tariff. In the earlier decades of the data, particularly from the 1950s through the late 1970s, the treatment and control groups diverged in their movements. During this period, the CPI index for new vehicles rises and falls much more sharply than the index for used vehicles. New vehicle prices experienced a steep upward trend beginning in the early 1970s, while used-vehicle prices increased more gradually. This indicates that the two groups were not evolving at the same

pace over the long run. From the 1980s through the pre-pandemic period, however, the two series move much more closely together. Both new and used vehicle prices exhibit similar trends, with comparable slopes and responses to broader macroeconomic events. For example, both series decline slightly during the Great Financial Crisis of 2008–2009, then resume a gradual upward trend afterward. This closer movement between the data during the decades immediately preceding the treatment period provides more support for the parallel trends assumption relevant to the Difference-in-Differences design.

Despite the strength of this empirical strategy, several limitations exist. Used vehicles may not perfectly represent the price trend for new vehicles in the absence of tariffs. New and used vehicle markets differ structurally, which may cause the two groups to respond to broader domestic economic shocks differently. Since both groups exist within the larger automobile market, shocks affecting new vehicles may also indirectly affect used vehicle prices. As a result, the estimated coefficient on tariffs may understate the true extent of tariff pass-through into new vehicle prices. The pandemic and subsequent supply chain disruptions created unusually large shocks within automobile markets that may continue to influence pricing in the post-treatment period despite the use of seasonally adjusted CPI data. Additionally, because the tariffs were implemented relatively recently, the post-treatment period remains short. As a result, the estimates may capture relatively short-run price responses.

4. Results and Discussion

Dependent Variable: Model:	(1)	log_price (2)
<i>Variables</i>		
Constant	4.361*** (0.0253)	
treat	0.1969*** (0.0299)	0.1969*** (0.0098)
post	0.8516*** (0.0256)	
treatpost	-0.2240*** (0.0301)	-0.2240*** (0.0106)
<i>Fixed-effects</i>		
date		Yes
<i>Fit statistics</i>		
Observations	1,758	1,758
R ²	0.04111	0.94786
Within R ²		0.31607
<i>Heteroskedasticity-robust standard-errors in parentheses</i>		
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>		

Examining the regression results, the period following the implementation of the 25% tariff on imported vehicles and automobile parts is associated with a 22.4% decrease in the logged CPI price index for new vehicles. These results are consistent across the difference-in-differences model with robust standard errors that control for fixed time effects and the model without such controls. Both models use heteroskedasticity-robust standard errors to mitigate the influence of heteroskedasticity in the data. In both regressions, the coefficient is significant at the 99% confidence level, indicating that it differs statistically from zero (i.e., the null can be rejected). In the second regression, the post-period coefficient disappears because the inclusion of time fixed effects absorbs the post-tariff trend common to both groups. Notably, the R² increases from 0.04111 in the first regression to 0.94786. The addition of fixed time effects in the second regression increased the explained variation by a linear model from 4.11% to 94.79%, leaving about 4% of the variance unexplained, overall strengthening the predictive accuracy of the model.

These results are inconsistent with much of the existing literature on tariff pass-through, which generally finds near-complete pass-through of tariffs into domestic prices. Based on prior research, one would expect tariffs on imported automobiles and automobile parts to increase the prices received by consumers. Instead, the empirical results suggest little pass-through of tariffs into new vehicle CPI prices during the post-tariff period. This finding implies that domestic manufacturers or importers may have absorbed part of the tariff burden rather than fully passing on increased costs to consumers through higher prices.

This misalignment of findings relative to other relevant literature suggests the presence of an additional, hidden mechanism affecting prices in the automobile market. One possible explanation stems from consumer choice theory, particularly the substitution effect. The substitution effect refers to consumers' tendency to substitute away from relatively more expensive goods towards cheaper alternatives when the price of a good rises, while income remains unchanged. Assuming tariffs increased the costs of new vehicles, consumers may have shifted demand toward used vehicles as relatively cheaper substitutes. The increased demand in the used vehicle market weakens demand for new vehicles while simultaneously pushing up used vehicle prices. The reduced demand for new vehicles may lead importers or domestic manufacturers to absorb part of the tariff burden through lower margins or financing incentives, as in the post-pandemic period, to maintain sales volumes. Under this interpretation, tariffs may still increase costs while generating weaker pass-through into the prices of tariff-affected goods due to substitution within the broader industry.

5. Conclusion

This paper examined the extent to which the 2025 tariff on imported automobiles and automobile parts was passed through into prices received by U.S. consumers. Using seasonally adjusted Consumer Price Index (CPI) data published by the Bureau of Labor Statistics for new vehicles and used cars and trucks from January 1953

through March 2026, this paper employed a difference-in-differences model to estimate the effect of tariffs on vehicle prices. This empirical strategy compared changes in the logged CPI price index for new vehicles relative to used vehicles following the implementation of the tariffs. Contrary to the initial hypothesis and much of the existing literature on tariff pass-through, the results suggest that prices for new vehicles declined following the implementation of the tariffs. These findings suggest little pass-through of tariffs into new vehicle CPI prices post-tariffs. This paper argues that a possible explanation for the observed findings is the presence of the substitution effect in the automobile industry, where consumers shifted demand away from new vehicles and toward used vehicles in response to tariff-induced price increases. Under this interpretation, tariffs may still increase costs while generating minimal pass-through into consumer prices. These findings contribute to the growing literature examining the domestic effects of tariffs and demonstrate the importance of examining industry-level consumer price responses rather than relying solely on aggregate inflation measures. More broadly, the results suggest that tariffs may affect markets through multiple channels beyond direct price increases, particularly in industries where consumers have access to close substitute goods.

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